

BÁO CÁO THỰC HÀNH

Thực hành Nhập môn Mạng máy tính

Lab 01: Mật mã học Cổ điển

GVTH: Nguyễn Ngọc Trưởng

Ngày báo cáo: 05/03/2026

Nhóm 14 – NT101.Q21.1

THÔNG TIN CHUNG

MSSV	Họ và tên	Nội dung báo cáo	% công việc
24520358	Lê Quốc Dương	Câu 2.1, câu 2.2	33.33%
24520365	Đặng Trần Khánh Duy	Câu 2.4, câu 2.5	33.33%
24521250	Huỳnh Vũ Minh Nhật	Câu 2.6, câu 2.7	33.33%

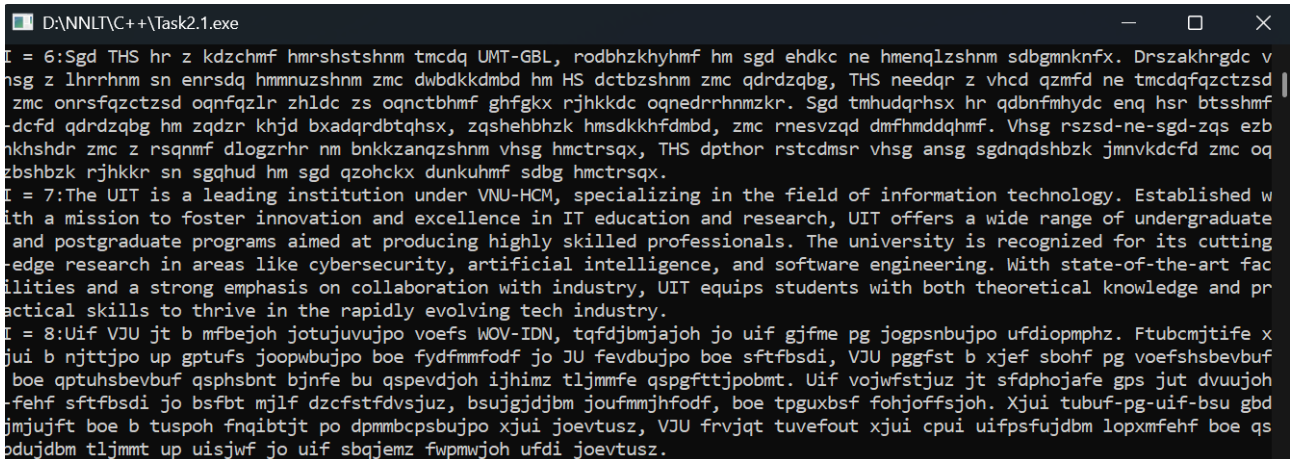
Task 2.1 :

Bạn hãy viết một chương trình bằng ngôn ngữ lập trình tự chọn để thực hiện mã hóa và giải mã bằng Caesar cipher.

1. Cho phép người dùng nhập một khóa và một chuỗi văn bản để thực hiện mã hóa hoặc giải mã.
2. Thực hiện brute-force để tìm bản rõ khi chỉ biết ciphertext và hiển thị kết quả giải mã duy nhất và khóa tương ứng. Sau đó, hãy sử dụng chương trình của bạn để giải mã đoạn văn bản mã hóa sau: (<https://pastecode.io/s/y3vj3cqe>) Max NBM bl t extwbgz bglmbmnmbhg ngwxk OGN-AVF, lixvbtebsbgz bg max ybxew hy bgyhkftmbhg mxvaghehxr. Xlmtueblaxw pbma t fbllbhg mh yhlmxk bgghotmbhg tgw xqvxeexgvx bg BM xwnvtmbhg tgw kxlxtkva, NBM hyyxkl t pbwx ktgzx hy ngwxkzktwntmx tgw ihlmzktwntmx ikhzkftl tbfw tm ikhwnvbgz abzaer ldbeexw ikhyxllbhgtel. Max ngboxklbmr bl kxvhzgsxw yhk bml vnmmbgz-xwzx kxlxtkva bg tkxtl ebdx vruxklxvnkbmr, tkmbbybvte bgmxeebxgvs, tgw lhymptkx xgzbxxkbgz. Pbma lmtmxhy-max-tkm ytvbebmbxl tgw t lmkhgz xfiatlbl hg vheetuhktmbhg pbma bgwnlmkr, NBM xjnbil lmnwxgml pbma uhma maxhkxmbvte dghpexwzx tgw iktvmbvte ldbeel mh makbox bg max ktibwer xoheobgz mxva bgwnlmkr.

Giải quyết : Kết quả sau khi giải mã thành công :

Với K = 7: The UIT is a leading institution under VNU-HCM, specializing in the field of information technology. Established with a mission to foster innovation and excellence in IT education and research, UIT offers a wide range of undergraduate and postgraduate programs aimed at producing highly skilled professionals. The university is recognized for its cutting-edge research in areas like cybersecurity, artificial intelligence, and software engineering. With state-of-the-art facilities and a strong emphasis on collaboration with industry, UIT equips students with both theoretical knowledge and practical skills to thrive in the rapidly evolving tech industry.



```
I = 6:Sgd THS hr z kdzchmf hmrshstshnm tmcddq UMT-GBL, rodbzhkzhyhmf hm sgd ehdkc ne hmenqlzshnm sdbgmknkfx. Drszakhrgcd v
ng z lhrhrnm sn enrsdq hmnuzshnm zmc dwbdkkdmdb hm HS dctbzshnm zmc qdrdzqbg, THS needqr z vhd qzmf ne tmcddqfzctzsd
zmc onrsfqzctzsd oqnfqzlr zhldc zs oqncbtbhmf ghfgkx rjhkkdc oqnedrrhnmzkr. Sgd tmhudqrhxs hr qdbnfmhydc enq hsr btsshmf
-dcdf qdrdzqbg hm zqdzr khjd bxadqrdbtqhsx, zqshebhzk hmsdkkhfmdmb, zmc rnesvzqd dmfmhddqhmfm. Vhsg rszsd-ne-sgd-zqs ezb
nkshdr zmc z rsqnmf dlogzrhr nm bnkzanzqshnm vhsg hmctrsqx, THS dpthor rstcdmsr vhsg ansg sgdngqshbzk jmnvkdcd zmc oq
zbshbzk rjhkkc sn sgghud hm sgd qzohckx dunkuhmf sdbg hmctrsqx.
I = 7:The UIT is a leading institution under VNU-HCM, specializing in the field of information technology. Established w
ith a mission to foster innovation and excellence in IT education and research, UIT offers a wide range of undergraduate
and postgraduate programs aimed at producing highly skilled professionals. The university is recognized for its cutting
-edge research in areas like cybersecurity, artificial intelligence, and software engineering. With state-of-the-art fac
ilities and a strong emphasis on collaboration with industry, UIT equips students with both theoretical knowledge and pr
actical skills to thrive in the rapidly evolving tech industry.
I = 8:Uif VJU jt b mfbejoh jotujuvujpo voefs WOV-IDN, tqfdjbmajoh jo uif gjfme pg jogpsnbujpo ufdiopmphz. Ftubcmjtife x
jui b njttjpo up gptufs joopwbujpo boe fydfmmfodf jo JU fevdubujpo boe sftfbsdi, VJU pggfst b xjef sbhof pg voefshsbevbuf
boe qptuhsbevbuf qspbsnt bnf bu qspvjdjoh ijhmz tljmmfe qspgfttjpbomt. Uif vojwfstjuz jt sfdphojafe gps jut dvuujoh
-fehf sftfbsdi jo bsfht mjlz dczfstfdvsjuz, bsujgdjbm jofummjhfof, boe tpguxbsf fohjoffsjoh. Xjui tubuf-pg-uif-bsu gbd
jmjujft boe b tuspoh fnqibtjt po dpmmbcpsbujpo xjui joevtusz, VJU frvjqat tuvefout xjui cpui uifpsfujdbm lopxmfehfe boe qs
odujdbm tljmmt up uisjwf jo uif sbqjemz fwpmwjoh ufdi joevtusz.
```

Task 2.2:

Bạn được cung cấp một tệp ciphertext (2.2_cipher_text) (<https://pastecode.io/s/h3hvp47p>) đã được mã hóa bằng mono-alphabetic substitution cipher. Nhiệm vụ là khôi phục lại bản rõ bằng phương pháp frequency analysis và mô tả chi tiết quy trình thực hiện theo từng bước, bao gồm cách thống kê tần suất, cách đưa ra giả thuyết về các ký tự, và cách kiểm tra, điều chỉnh cho đến khi thu được văn bản có nghĩa. Biết trước rằng bản rõ là một bài báo tiếng Anh.

Kết quả sau khi giải mã thành công : The Harry Potter series, written by British author J.K. Rowling, is one of the most popular and influential fantasy sagas of modern literature. Spanning seven books, the series follows the life of a young boy, Harry Potter, who discovers on his 11th birthday that he is a wizard. He is invited to attend Hogwarts School of Witchcraft and Wizardry, where he learns about his magical heritage, develops close friendships, and confronts the dark forces that threaten the wizarding world.

The central plot revolves around Harry's ongoing battle with Lord Voldemort, a dark wizard intent on conquering both the magical and non-magical worlds. As Harry grows older, he learns more about his own mysterious past, his connection to Voldemort, and the prophecy that ties their fates together. Throughout the series, themes of friendship, loyalty, bravery, and the power of love are explored, with a rich world filled with magical creatures, spells, and deep mythology.

The books, starting with Harry Potter and the Philosopher's Stone (1997) and concluding with Harry Potter and the Deathly Hallows (2007), have sold over 500 million copies worldwide, been translated into more than 80 languages, and adapted into a successful film series. The series is known not only for its enchanting world-building but also for addressing darker, complex themes such as mortality, prejudice, and the abuse of power. The Harry Potter series has left an indelible mark on popular

culture, fostering a global fan base, inspiring countless spin-offs, and cementing its place as a literary phenomenon.

Vietnam National University, Ho Chi Minh City (VNU-HCM) is one of the most prestigious and comprehensive higher education institutions in Vietnam, known for its excellence in research, innovation, and education. Founded in 1995 by the Vietnamese government, VNU-HCM was established as a key center for training, research, and technology transfer with the mission of contributing to the country's socio-economic development. As a flagship university, it has played a pivotal role in producing highly skilled professionals, scientists, and leaders across multiple fields. VNU-HCM operates under a distinctive structure as a multi-disciplinary and multi-campus system, comprising several member universities and research institutes. These include some of the most well-established institutions in the region, such as the University of Science, University of Technology, University of Social Sciences and Humanities, International University, University of Economics and Law, and others. Each of these member institutions operates semi-independently but collaborates under the umbrella of VNU-HCM, making it a unique higher education model in Vietnam that promotes interdisciplinary learning and research.

Located in Ho Chi Minh City, the country's economic and cultural hub, VNU-HCM benefits from close ties with industries, businesses, and government agencies. This proximity fosters collaboration in research, internships, and job placement, ensuring that its graduates are well-prepared to meet the demands of an evolving workforce. The university is committed to producing graduates who are not only academically proficient but also equipped with practical skills and a global outlook.

VNU-HCM is renowned for its strong emphasis on scientific research and technological innovation. It houses numerous research centers, institutes, and labs that focus on various areas of science, engineering, social sciences, economics, and more. The university's research agenda is aligned with national priorities, focusing on areas such as information technology, biotechnology, renewable energy, and environmental sustainability. VNU-HCM also actively engages in international cooperation, partnering with leading universities and research institutions worldwide to enhance its academic programs, foster joint research projects, and promote student and faculty exchanges.

The university is also deeply involved in addressing global challenges through its research and educational programs. VNU-HCM places a strong emphasis on sustainable development, environmental conservation, and technological innovation, aligning its goals with the United Nations' Sustainable Development Goals (SDGs).

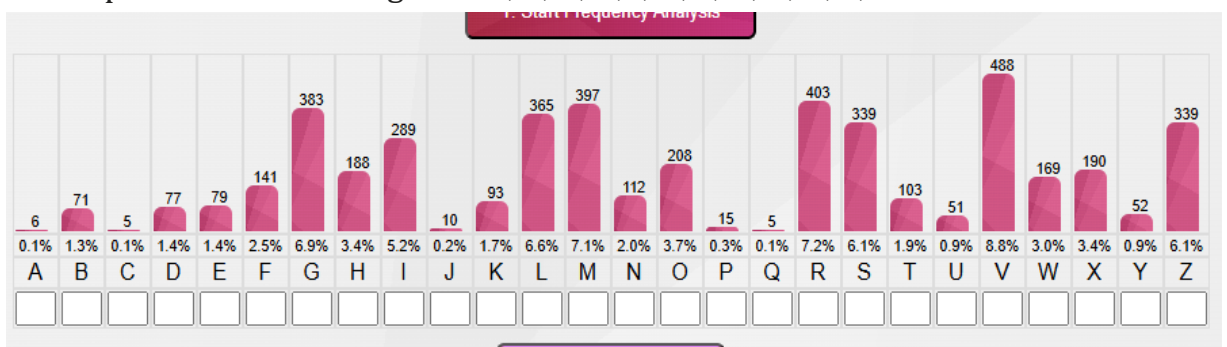
The institution encourages its faculty and students to pursue research that has a meaningful impact on both the local and global community.

In terms of student life, VNU-HCM offers a vibrant and dynamic environment. With a student body of over 60,000 and a faculty of highly qualified professors and researchers, the university provides a rich academic and social experience. The campuses are equipped with modern facilities, including libraries, laboratories, sports complexes, and dormitories, creating a supportive learning environment. Students also have access to a variety of extracurricular activities, clubs, and organizations, allowing them to engage in cultural, social, and professional development alongside their studies.

In conclusion, Vietnam National University - Ho Chi Minh City stands as a leading educational institution in Vietnam, contributing significantly to the country's intellectual, scientific, and economic progress. Its commitment to academic excellence, research innovation, and societal impact has positioned it as a crucial player in shaping the future of Vietnam and the broader region. As it continues to evolve and expand, VNU-HCM remains dedicated to fostering the next generation of leaders and innovators who will drive sustainable growth and global progress.

Cách thức thực hiện giải mã :

- Chuẩn hóa dữ liệu đầu vào
 - Bỏ qua khoảng trắng, dấu câu khi thống kê.
 - Chuyển toàn bộ chữ về cùng một kiểu (thường là uppercase) để đếm ổn định.
- Thống kê tần suất ký tự đơn
 - Đếm số lần xuất hiện của 26 chữ cái trong ciphertext.
 - So với phân bố chuẩn tiếng Anh: E, T, A, O, I, N, S, H, R, D, L, U...



- Tạo giả thuyết ánh xạ ban đầu
 - Chữ xuất hiện nhiều nhất trong mã thường thử map với E hoặc T trước.
 - Với đoạn này, cụm Ghv xuất hiện rất nhiều ở đầu câu, giả thuyết mạnh là The.
 - Từ đó suy ra ngay: g -> t, h -> h, v -> e.
- Dùng mẫu từ ngắn để khóa thêm mapping
 - rs -> is => r -> i, s -> s

- lmw -> and => l -> a, m -> n, w -> d
- yb -> by => y -> b, b -> y
- zmv -> one => z -> o, m -> n, v -> e

Các từ cực ngắn này giúp giảm sai số rất nhanh.

5. Dùng từ riêng và ngữ cảnh để xác nhận

- Hlib Kzggvi khớp Harry Potter
 - Q.P. Izuormt khớp J.K. Rowling
- Khi tên riêng khớp, mapping gần như chắc chắn đúng.

6. Mở rộng mapping theo cụm dài

- Giải từng câu, chỗ nào ra từ “gần đúng” thì chỉnh một vài ký tự chưa chắc.
- Ví dụ kiểu _nfl_ential, _iterature, _niversity... giúp chốt các chữ hiếm.

7. Lặp kiểm tra cho đến khi văn bản “trôi chảy”

- Tiêu chí dừng: câu tiếng Anh có ngữ pháp, dấu câu tự nhiên, thuật ngữ nhất quán.
- Đến cuối, toàn văn bản đọc trơn tru, nghĩa hoàn chỉnh như một bài báo/giới thiệu.

Cipher->plain

- a -> q
- b -> y
- c -> z
- d -> f
- e -> v
- f -> u
- g -> t
- h -> h
- i -> r
- j -> x
- k -> p
- l -> a
- m -> n
- n -> m
- o -> l
- p -> k
- q -> j
- r -> i
- s -> s
- t -> g
- u -> w
- v -> e

- w -> d
- x -> c
- y -> b
- z -> o

Task 2.4: Bạn hãy tìm hiểu thuật toán mã hóa và giải mã Playfair Cipher và viết một chương trình để mã hóa và giải mã bằng Playfair cipher. 1. Cho phép nhập khóa và bản rõ để mã hóa hoặc nhập khóa và ciphertext để giải mã. (Lưu ý: Chương trình phải hiển thị ma trận Playfair 5×5 tương ứng 5 với khóa đã nhập.) 2. Sau khi hoàn thành, sinh viên kiểm thử chương trình với một văn bản dưới đây với khóa được sử

dụng là HarryPotter.(<https://pastecode.io/s/c9oktc3h>)

ARYWYPHCBVEBYGMPNCYGCNTDNCWTMGRMFTQP
LEWTMLREFBEBQEBIYGBFLPHVOAEHKDHEUNGQ
FEROLEWTMLOPHEQGOSBEROQDWTLTCLMTHBWLNR
KXRYLORYYPHCBVEBYRLGYDMKYGGWKLROANDB
WGNERMNGYRLGHEWTRLMBRHMUDGVODVTEGMC
HLGWCMTFODNRRYCMZKODDUTDXGEOPOYRMFRM
GUKXRYGHABROVTGQMCEHPRPEOTSEGEQLARYW
YPOTMGQDOEXGOAUDHGUTULTNEHFTFHPGXGVP
HGURBDMEGWKLETCBOTNTFQLTAEHMTUGEAOAHE
VEROXGVPHGDEWTEWGQIEDLPILERWPMOATNGQ

KQEAHBMVRFKBRMKLXODXFREBHMNUKXRYKLRLM FLWDDNCN

Giải Quyết :

Harry Potter is a series of seven fantasy novels written by British author J.K. Rowling. The novels chronicle the lives of a young wizard, Harry Potter, and his friends Hermione Granger and Ron Weasley, all of whom are students at Hogwarts School of Witchcraft and Wizardry.

The main story arc concerns Harry's conflict with Lord Voldemort, a dark wizard who intends to become immortal, overthrow the wizard governing body known as the Ministry of Magic, and subjugate all wizards and Muggles.


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D:\NNLTC+ \Task2.4.exe
HLGWCMTFODNRRYCMZKODDUTDXGEOPOYRMFRM
GUKXRYGHABROVTGQMCHEPRPEOTSEGEQLARYW
YPOTMGQDOEXGOAUDHGUTULTNEHFTFHPGXGVP
HGURBDMEGWKLCTBOTNTFQLTAEHMTUGEOAHE
VEROXGVPHGDEWTEWQIEDLPILERWPMOATNGQ
KQEAHBMVRFKBRMKLXODXFREBHMNUKXRYKLRM
FLWDDNCN
END

Cipher normalized:
AR YW YP HC BV EB YG MP NC YG CN TD NC WT MG RM FT QP LE WT ML RE FB EB QE BI YG BF LP HV OA EH KD HE UN GQ FE RO LE WT ML
OP HE QG OS BE RO QD WT LC MT HB WL NR KX RY LO RY YP HC BV EB YR LG YD MK YG GW KL RO AN DB WG NE RM NG YR LG HE WR TR L
M BR HM UD GV OD VT EG MC HL GW CM TF OD NR RY CM ZK OD DU TD XG EO PO YR MF RM GU KX RY GH AB RO VT GQ MC EH PR PE OT SE
GE QL AR YW YP OT MG QD OE XG OA UD HG UT UL TN EH FT FH PG XG VP HG UR BD ME GW KL ET CB OT NT FQ LT AE HM TU GE OA HE VE
RO XG VP HG DE WT EW GQ IE DL PI LE RW PM OA TN GQ KQ EA HB MV RF KB RM KL XO DX FR EB HM NU KX RY KL RM FL WD DN CN

Plain RAW:
HA RX RY PO TX TE RI SA SE RI ES OF SE VE NF AN TA SY NO VE LS WR IT TE NB YB RI TI SH AU TH OR IK RO WL IN GT HE NO VE LS
CH RO NI CL ET HE LI VE SO FA YO UN GW IZ AR DH AR RY PO TX TE RA ND HI SF RI EN DS HE RM IO NE GR AN GE RA ND RO NW EA S
L EY AL LO FW HO MA RE ST UD EN TS AT HO GW AR TS SC HO OL OF WI TC HC RA FT AN DW IZ AR DR YT HE MA IN ST OR YA RC CO NC
ER NS HA RX RY CO NF LI CT WI TH LO RD VO LD EM OR TA DA RK WI ZA RD WH OI NT EN DS TO BE CO ME IM MO RT AL OV ER TH RO WT
HE WI ZA RD GO VE RN IN GB OD YK NO WN AS TH EM IN IS TR YO FM AG IC AN DS UB IU GA TE AL LW IZ AR DS AN DM UG GL ES

Plain CLEAN:
HA RR YP OT TE RI SA SE RI ES OF SE VE NF AN TA SY NO VE LS WR IT TE NB YB RI TI SH AU TH OR IK RO WL IN GT HE NO VE LS CH
RO NI CL ET HE LI VE SO FA YO UN GW IZ AR DH AR RY PO TT ER AN DH IS FR IE ND SH ER MI ON EG RA NG ER AN DR ON WE AS LE Y
A LL OF WH OM AR ES TU DE NT SA TH OG WA RT SS CH OO LO FW IT CH CR AF TA ND WI ZA RD RY TH EM AI NS TO RY AR CC ON CE RN
SH AR RY CO NF LI CT WI TH LO RD VO LD EM OR TA DA RK WI ZA RD WH OI NT EN DS TO BE CO ME IM MO RT AL OV ER TH RO WT HE WI
ZA RD GO VE RN IN GB OD YK NO WN AS TH EM IN IS TR YO FM AG IC AN DS UB IU GA TE AL LW IZ AR DS AN DM UG GL ES

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Task 2.5: Bạn hãy viết một chương trình bằng ngôn ngữ lập trình tự chọn để thực hiện mã hóa và giải mã bằng Vigenère cipher. Chương trình cho phép nhập khóa và bản rõ để mã hóa hoặc nhập khóa và ciphertext để giải mã. Sau khi hoàn thành, sinh viên kiểm thử chương trình với một văn bản có độ dài tối thiểu 100 từ, thực hiện mã hóa và giải mã lại để kiểm tra tính đúng đắn, đồng thời so sánh kết quả với các công cụ online khác

Giải Quyết : Key : LEQUOCDUONG

Encrypt: Life in the city is full of activity. Early in the morning, people leave their houses in the manner ants do when their nest is broken. Then the streets are full of traffic. Shops and offices open, students are eager to go to their school and the day's work begins. Many sightseers and tourists also visit many tourist attractions in the city, and businessmen from many parts of the world also come to conduct business transactions. In the evening, the offices and day schools begin to close, many of the shops are too close. There is often a traffic jam at this time because everyone wants to get home quickly. The city is described as a place of constant activity.


```

D:\NNLT\C++\Task2.5.exe
Nhap text: Life in the city is full of activity. Early in the morning, people leave their houses in the manner ants do when their nest is broken. Then the streets are full of traffic. Shops and offices open, students are eager to go to their school and the day's work begins. Many sightseers and tourists also visit many tourist attractions in the city, and businessmen from many parts of the world also come to conduct business transactions. In the evening, the offices and day schools begin to close, many of the shops are too close. There is often a traffic jam at this time because everyone wants to get home quickly. The city is described as a place of constant activity.
Nhap key: LEQUOCDUONG
Chon che do (E = Encrypt, D = Decrypt): E
Ciphertext: WMVYWPWBSPOECYMTWOFCSGNXYPWVBYOERJMDNVGPIFAOYKFYCROYZRGGIJBKUBCHYPWYHHJHGOATPVQHHUGIKUKYXXYWTQYGGODFHIYGQNV
RTELUMHTHYHFGCIVOZNRZHEGQJYWGJRJGNTOSVZWEHMCCKYWJORGQNGNXPIQASTWIUBZXXXYWTVWVBUWEDXHJHXOLYHSHEPGJCBFSLROMWIKNGRKCWQHRVRO
FVYEWQFGQYCGVZXDSDSHQXLWFLXJLOEWCCAYTRJBSELNMNTOFKMWPHMGZKYJHIAODHMCXCIIITVKYKBXWHQFGQFIARZZGEHRWFNPHYTRUMGVUUBFGNXYIBUL
HHUKPZUHPJNVRUQJYWSUDHRQGWJSBCQOMPRMTRJIGNRMSZGYCEZHJHMBVDEHYHQRWZBYPXXYFGLMCSZPRQNFICIZWPPLQQNHJLMHVSPFUWOWVYSIKCCEHSY
DHHFZZKUNVQPYEHONOBESHJHWWGETWTYGEUCPRJLWQJZCFYCSIZRINOPWUQGOGMJS
Process returned 0 (0x0)   execution time : 12.632 s

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Decrypt:

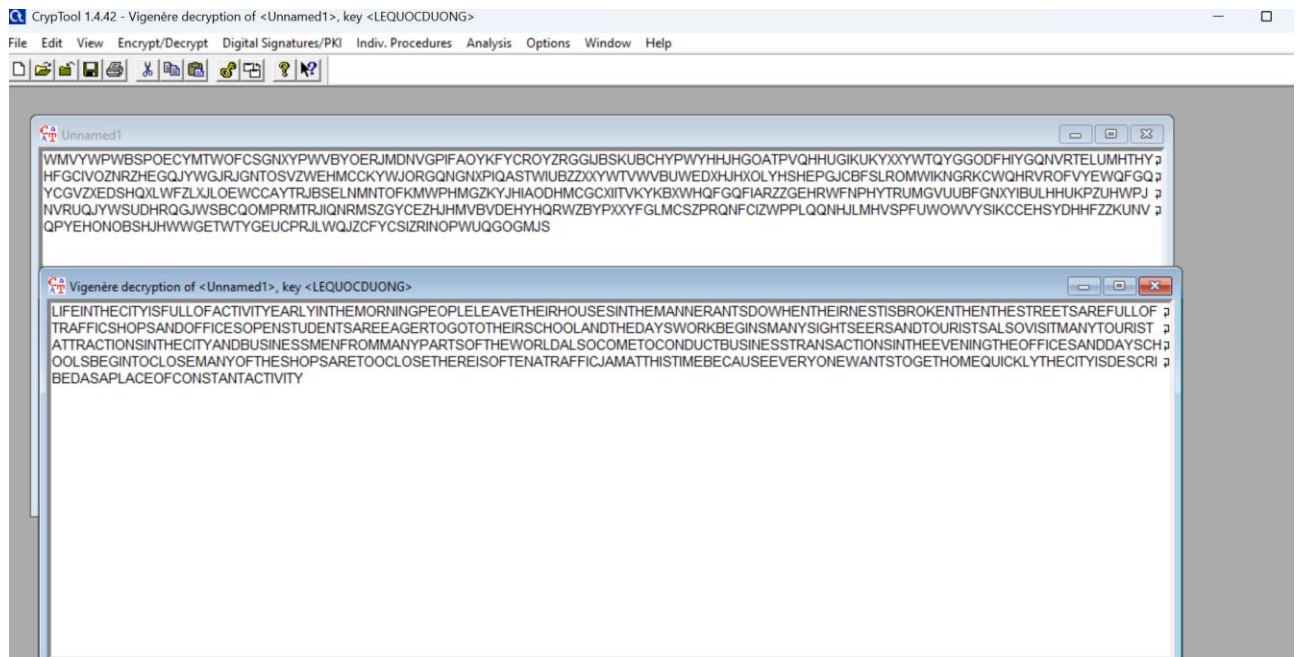
WMVYWPWBSPOECYMTWOFCSGNXYPWVBYOERJMDNVGPIFAOYKFYCROYZRGGIJBKUBCHYPWYHHJHGOATPVQHHUGIKUKYXXYWTQYGGODFHIYGQNV RTELUMHTHYHFGCIVOZNRZHEGQJYWGJRJGNTOSVZWEHMCCKYWJORGQNGNXPIQASTWIUBZXXXYWTVWVBUWEDXHJHXOLYHSHEPGJCBFSLROMWIKNGRKCWQHRVROFVYEWQFGQYCGVZXDSDSHQXLWFLXJLOEWCCAYTRJBSELNMNTOFKMWPHMGZKYJHIAODHMCXCIIITVKYKBXWHQFGQFIARZZGEHRWFNPHYTRUMGVUUBFGNXYIBULHHUKPZUHPJNVRUQJYWSUDHRQGWJSBCQOMPRMTRJIGNRMSZGYCEZHJHMBVDEHYHQRWZBYPXXYFGLMCSZPRQNFICIZWPPLQQNHJLMHVSPFUWOWVYSIKCCEHSY DHHFZZKUNVQPYEHONOBESHJHWWGETWTYGEUCPRJLWQJZCFYCSIZRINOPWUQGOGMJS

```

D:\NNLT\C++\Task2.5.exe
Nhap text: WMVYWPWBSPOECYMTWOFCSGNXYPWVBYOERJMDNVGPIFAOYKFYCROYZRGGIJBKUBCHYPWYHHJHGOATPVQHHUGIKUKYXXYWTQYGGODFHIYGQNV
RTELUMHTHYHFGCIVOZNRZHEGQJYWGJRJGNTOSVZWEHMCCKYWJORGQNGNXPIQASTWIUBZXXXYWTVWVBUWEDXHJHXOLYHSHEPGJCBFSLROMWIKNGRKCWQHRVRO
FVYEWQFGQYCGVZXDSDSHQXLWFLXJLOEWCCAYTRJBSELNMNTOFKMWPHMGZKYJHIAODHMCXCIIITVKYKBXWHQFGQFIARZZGEHRWFNPHYTRUMGVUUBFGNXYIBUL
HHUKPZUHPJNVRUQJYWSUDHRQGWJSBCQOMPRMTRJIGNRMSZGYCEZHJHMBVDEHYHQRWZBYPXXYFGLMCSZPRQNFICIZWPPLQQNHJLMHVSPFUWOWVYSIKCCEHSY
DHHFZZKUNVQPYEHONOBESHJHWWGETWTYGEUCPRJLWQJZCFYCSIZRINOPWUQGOGMJS
Nhap key: LEQUOCDUONG
Chon che do (E = Encrypt, D = Decrypt): D
Plaintext: LIFEINTHECITYISFULLOFACTIVITYEARLYINTHEMORNINGPEOPLELEAVETHEIRHOUSESINTHEMANNERANTSDOWHENTHEIRNESTISBROKENTH
INTHESTREETSAREFULLOFTRAFFICSHOPSANDOFFICESOPENSTUDENTSAREEAGERTOGOTOTHEIRSCHOOLANDTHEDAYSWORKBEGINSMANYSIGHTSEERSANDTOU
ISTSALSOVISITMANYTOURISTATTRACTIONSINTHECITYANDBUSINESSMENFROMMANYPARTSOFTHEWORLDALSOOMETOCONDUCTBUSINESSTRANSACTIONSIN
THEEVENINGTHEOFFICESANDDAYSSCHOOLSBEGINTOCLOSEMANYOFTHESHOPSARETOOCLOSETHEREISOFTENATRAFFICJAMATTHISTIMEBECAUSEEVERYONEW
NTSTOGETHOMEQUICKLYTHECITYISDESCRIBEDASAPLACEOFCONSTANTACTIVITY

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Kiểm Tra :



Task 2.6:

Bạn hãy xây dựng một chương trình hỗ trợ phá mã Vigenère cipher khi không biết khóa. Chương trình cần cho phép nhập hoặc đọc từ file một ciphertext bất kỳ và thực hiện quá trình phân tích để tự động tìm khóa và bản rõ có khả năng đúng cao nhất. Kết quả cần hiển thị khóa tìm được, bản rõ tương ứng và một số thông tin cơ bản về quá trình phân tích. Sinh viên sử dụng chương trình để giải mã ciphertext được cung cấp và trình bày kết quả. (<https://pastecode.io/s/wm8dsq5z>) pp oiuibvql avpgzwm, vyabnwyzcbbg klhqla mv uqwckl kzzwktcfcwg hpp wwftwzcktakah bxjjzcynlu pyzbcgp zzht onmpxtcfckts eahkxwve uvw h uqn wy ywxy-jtzgp wiejwxubbvpe wiesgp utzvtunpfz va nztuurizf tgemizlu uh etfu fbim htq bikk va xmvprtyz. mogey lxagdgqgpufck tsialqmooe uzx buqx nhy edsxmviduxape wyg zlpqlimpqz uvw kkscbts uuavbui mhl oltuzqvhvui mv rdibxjv pubt wtupivf, yqv jkvyecvz vp fbm buvqlvxa czx khuhuxmgakmf khtoghqvhvui vl zwob il jtqxqm jcdx bkhpeukmpqzm igk gyuqe

Giải Quyết :

```

Nhap ban ma (Xuong dong 2 lan de bat dau):
pp oiuibvql avpgzwm, vyabnwzycbbg klhqla mv uqwk1 kzzwktcfwg hpp
mwftwzckதாக bxjjzcnlu pyzbcgp zzht omnpxtcfckts eahkxwve uvw
h uqn wy ywxy-jtzgp wiejwxubbvpe wiesgp utzvtunpfz va nztuurizf
tgemizlu uh etfu fbim htq bikk va xmvprtyz. mogey lxagdgqgufck
tsialqmooe uzx buqx nhy edsxmviduxape wyg zlpqlimpqz uvw kkscbts
uuavbui mhl oltuzqvhvui v mv rdibxjv pubt wtupivf, yqv jkvyecvz vp
fbm buvqlvxa czx khuhuxmgakmf khtoghqvhvui vl zwob il jtqxqm jcdx
bkhpeukmpqzm igk gyuqe

=> Do dai khoa tim duoc: 6
=> Khoa tim duoc: HCMUIT

--- BAN RO ---
in computer science, cryptography refers to secure information and communication techniques derived from mathematical co
ncepts and a set of rule-based calculations called algorithms to transform messages in ways that are hard to decipher. t
hese deterministic algorithms are used for cryptographic key generation and digital signing and verification to protect
data privacy, web browsing on the internet and confidential communications such as credit card transactions and email
Process returned 0 (0x0)   execution time : 8.849 s
Press any key to continue.

```

Task 2.7:

Sinh viên lựa chọn một thuật toán mật mã cổ điển bất kỳ, trình bày ngắn gọn nguyên lý hoạt động và xây dựng một chương trình để thực hiện mã hóa và giải mã theo thuật toán đã chọn. Cần chuẩn bị một ví dụ minh họa để chứng minh chương trình hoạt động đúng, có thể bằng cách mã hóa và giải mã một đoạn văn bản mẫu.

Giải thuật Rail Fence Cipher (Zigzag Transposition)

1) Bản chất của Rail Fence

- Đây là **mã hoán vị (transposition cipher)**: không thay đổi chữ cái (không biến A thành D như Caesar), mà chỉ **đổi vị trí** các ký tự trong chuỗi.
- Khóa của nó là **số hàng (rails)**: ký hiệu là (r).

2) Nguyên lý mã hóa (Encryption)

Bước A — Vẽ “đường đi zigzag”

- Bạn tưởng tượng có (r) dòng (hàng) nằm ngang.
- Ta viết ký tự theo dạng zigzag:
 - + Bắt đầu ở hàng 0 (trên cùng).
 - + Đi xuống từng hàng cho đến hàng (r-1).
 - + Sau đó đổi chiều đi lên từng hàng cho đến hàng 0.
 - + Lặp lại như thế cho đến hết chuỗi.

Nói theo “trạng thái”:

Bước B — Đọc theo từng hàng để tạo ciphertext

- Sau khi “rải” hết ký tự thành zigzag trên các hàng,
- Ta tạo ciphertext bằng cách **đọc hàng 0 trước, rồi hàng 1, ... đến hàng r-1.**

Ví dụ trực quan

Plaintext: WEAREDISCOVEREDFLEEATONCE, (r=3)

Viết zigzag:

- Hàng 0: W ... E ... C ... R ... L ... T ... E

- Hàng 1: . E . R . D . S . O . E . E . F . E . A . O . C .
- Hàng 2: ... A ... I ... V ... D ... E ... N ...

Đọc theo hàng:

- Row0: WECRLTE
- Row1: ERDSOEEFEAOC
- Row2: AIVDEN

Ciphertext: WECRLTEERDSOEEFEAOCAIVDEN

3) Nguyên lý giải mã (Decryption)

Bước A — Biết trước “ô nào được điền”

- Với độ dài ciphertext là (n) và (r) rails, ta mô phỏng lại đường zigzag **nhưng chỉ đánh dấu** vị trí (cột nào thuộc hàng nào).
- Ta tạo một bảng (r \times n) (hoặc cấu trúc tương đương), tại mỗi cột chỉ có đúng 1 ô là “được đi qua”.

Ví dụ: nếu zigzag đi qua hàng 1 tại cột 5, ta đánh dấu ô (row=1, col=5).

Bước B — Điền ciphertext vào các ô theo thứ tự từng hàng

- Ciphertext được tạo bằng cách đọc theo hàng, nên khi giải mã:
 - + Ta đi theo hàng 0 từ trái qua phải, ô nào được đánh dấu thì điền ký tự ciphertext tiếp theo.
 - + Xong hàng 0 → sang hàng 1 → ... → hàng (r-1).

Bước C : Đọc lại theo đường zigzag để ra plaintext

- Sau khi bảng đã được điền đủ,
- Ta “đi theo zigzag” (giống lúc mã hóa) và đọc ký tự tại các ô tương ứng, sẽ thu lại plaintext.

Có 3 chế độ Encrypt, Decrypt và dò Rails bằng Brute-force

Plaint Text : Life in the city is full of activity. Early in the morning, people leave their houses in the manner ants do when their nest is broken. Then the streets are full of traffic. Shops and offices open, students are eager to go to their school and the day's work begins. Many sightseers and tourists also visit many tourist attractions in the city, and businessmen from many parts of the world also come to conduct business transactions. In the evening, the offices and day schools begin to close, many of the shops are too close. There is often a traffic jam at this time because everyone wants to get home quickly. The city is described as a place of constant activity.

Rails : 13

Encrypt : L h nesfptd ru rr lcnrmveeoiloteetnens tfioe on .Muiorecfooltb niei heea eem dscnflf vhar terac nr atsaostihi mwdcu. fsgteoh ja roq c seu nmaem r her.s,eg hnntt sttn usstf eohpTi teyhusrft faioei aiitet e goleiy syt

yemeadinhoab ts sc so iioai c rlrenes t Scsa o g d n n,mahlnnoe n c . ituntc b
 nnstyn hth nsfhitetodesnaaaai sntsoei edslfaeofhaeekyeet iliehtstbe oofu
 ohabialmt asy ocsteh loorsfffc gltdc t vrnlo rha pfde cy g s tsne f scvtdos
 eotasew yi aa.hyiagpundnoTrlsoerts'khsotrndnpoco ae aoey ler
 bao.calcyettE,osioek el nah srtr iao ia ottsn,yh,ntcntt nt sptt iy ee he. uadt er
 oevscibsrsm rnig c ao iet Te iic.pswnfnsiweituteansmoamshav

```

D:\NNLT\C++\Task2.7.exe
=== Rail Fence Cipher ===
1) Encrypt
2) Decrypt
3) Brute-force rails (decrypt)
Chon (1/2/3): 1
Nhap text (co the nhieu dong, go END de ket thuc):
Life in the city is full of activity. Early in the morning, people leave their houses in the manner ants do when their nest is broken. Then the streets are full of traffic. Shops and offices open, students are eager to go to their school and the day's work begins. Many sightseers and tourists also visit many tourist attractions in the city, and businessmen from many parts of the world also come to conduct business transactions. In the evening, the offices and day schools begin to close, many of the shops are too close. There is often a traffic jam at this time because everyone wants to get home quickly. The city is described as a place of constant activity.
END
Nhap so rails: 13

=== CIPHERTEXT ===
L h nesfptd ru rr Icnsrmveeoloteetnens tfioe on .Muiorecfooltb niei heea eem dscnflf vhar terac nr atsaostihi mwdcu.
fsgteoh ja roq c seu nmaem r her.s, eg hnntt sttn usstf eohpTi teyhusrft faioei aiitet e goleiyy syt yemeadinhoab ts sc
so iioai c rlrenes t Scsa o g d n n,mahlnnoe n c . ituntc b nnstyn hth nsfhitetodesnaaaai sntsoei edslfaeofhaeekyeet i
iliehtstbe oofu ohabialmt asy ocsteh loorsfffc gltdc t vrnlo rha pfde cy g s tsne f scvtdos eotasew yi aa.hyiagpundno
Trlsoerts'khsotrndnpoco ae aoey ler bao.calcyettE,osioek el nah srtr iao ia ottsn,yh,ntcntt nt sptt iy ee he. uadt e
r oevscibsrsm rnig c ao iet Te iic.pswnfnsiweituteansmoamshav

Process returned 0 (0x0) execution time : 19.477 s
Press any key to continue.

```

Decrypt :


```

D:\NNLT\C++\Task2.7.exe
=== Rail Fence Cipher ===
1) Encrypt
2) Decrypt
3) Brute-force rails (decrypt)
Chon (1/2/3): 2
Nhap text (co the nhieu dong, go END de ket thuc):
L h nesfptd ru rr Icnstrmveeoiloteetnens tfioe on .Muiorecfooltb niei heea eem dscnflf vhar terac nr atsaostihi mwdcu.
fsgteoh ja roq c seu nmaem r her.s, eg hnnntt sttn usstf eohpTi teyhusrft faioei aiitet e goleyi syt yemeadinhoab ts sc
so iiooi c rlrenes t Scsa o g d n n,mahlnnoe n c . ituntc b nnstyn hth nsfhitetodesnaaaai sntsoei edslfaeofhaeekyeet i
iliehtstbe oofu ohabialmt asy ocsteh loorsffic gltdc t vrnl o rha pfde cy g s tsne f scvtdos eotasew yi aa.hyiagpundno
Trlsoerts'khsotrndnpoco ae aoey ler bao.calcyettE,osioek el nah srtr iao ia ottsn,yh,ntcntt nt sptt iy ee he. uadt e
r osevsicbrsm rnig c ao iet Te iic.pswnfnsiweiteansmoamshav
END
Nhap so rails: 13

=== PLAINTEXT ===
Life in the city is full of activity. Early in the morning, people leave their houses in the manner ants do when their nest is broken. Then the streets are full of traffic. Shops and offices open, students are eager to go to their school and the day's work begins. Many sightseers and tourists also visit many tourist attractions in the city, and businessmen from many parts of the world also come to conduct business transactions. In the evening, the offices and day schools begin to close, many of the shops are too close. There is often a traffic jam at this time because everyone wants to get home quickly. The city is described as a place of constant activity.

Process returned 0 (0x0) execution time : 11.101 s
Press any key to continue.

```

Bảng Brute-force :

```

D:\NNLT\C++\Task2.7.exe
s ts aorschaeinbkws t ittyny ssrey sonelr foilsmm chi aoc iisa ornattiu tnsoee.anaihhtasre tio y gm sild faet ceo
arcs tgos estaornpraho usalhonsprafni a adfaeTteiiniino vi oc e

--- rails = 11 ---
Lt snb onetgen an et ss oc n sotcusnhsas yo at tetfnssso tsto us inan ftfnni ihoeti e htiehoeioihafy cs b oh
sompani lp.tesasiTwnf di flcrtp vmhc c.pMut ittbisyad tre.uti yleao w .sstrhfodrsueoy ceneae dnsge ctreesool f ctcssfef
root nsayhneestat houl f aeet.s twhaSajt baici Euia,y l so rnreasoiodwts on ioireq otrai eeoass a i cigof.eri khfe i
Ihstdiiy otseic eeceutne aleuv ggd an nenslp stcnutl, meseagmfdnaiebhdcacaoemr mlhe n sarsotole dmsrinf Trsnmtrvhny c
vnh oarlr sr sneesefelry eloinmiaeknt.fe s y rogo t ecy, ovee e sica a'r t mghia e. hk amoo hai ahlr ndii st s topltint
oetnuifts hinrdenhtrta othen,eatyd tcasce tb scn h VT,pytb tnernt

--- rails = 12 ---
Lt sinshe i bo o,y,te eee meltasipoaoiasgrthna yh os.tctcerelthce n n ndswa .il fsynnsn rteolfycyp yiifemt tefa e ac e
seta enastisostd ogatew E.lfcitaef onn hth, nhoyda h sopnt o.ecisef.ia e a i t.hubiktgo nuep yt siMdu stuevukasd nrenst
mi owf ttndeti lnl cs drrecec io o ewr T ib oucuf.hs l rn eoal inop o ofT nershsie ohhsiiislrte ittaesvtsrr syittbr eeontp
ttcuirsfb ayon ihhi ed' btsike h u eIijscto haresoscoh ra c fr fyo ma it unltrhneo rs grarnn n ofefqesa anhsdoisgtn ai
eic re ostbtpt mcsosie esemmeittancn oa,oelo u vd aSdm yoa haftecinsecmiss e, m n anatanefee assatisecocyaotmlof a
venth tytoiger i cd tTae lossdghvlheo ntsen v tr etnlraor.e

--- rails = 13 ---
Life in the city is full of activity. Early in the morning, people leave their houses in the manner ants do when their nest is broken. Then the streets are full of traffic. Shops and offices open, students are eager to go to their school and the day's work begins. Many sightseers and tourists also visit many tourist attractions in the city, and businessmen from many parts of the world also come to conduct business transactions. In the evening, the offices and day schools begin to close, many of the shops are too close. There is often a traffic jam at this time because everyone wants to get home quickly. The city is described as a place of constant activity.

Process returned 0 (0x0) execution time : 10.168 s

```

Link github : <https://github.com/lequocduong1207/NT101.git>

HẾT